

# Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU  
FIFTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Embedded Systems [ECM\_3122]

Marks: 50

Duration: 180 mins.

### PART-A

Answer all the questions.

- 1) Differentiate between i) General purpose systems and Embedded Systems. ii) RISC and CISC. (4)  
A) Course Outcome: "CO1" / Cognitive Level: "Understand"  
B) Elaborate on the mechanism of Bus technology situated in ARM processors. (3)  
Course Outcome: "CO2" / Cognitive Level: "Application"  
C) Expand the following with syntax: (3)  
i) AREA ii) ENTRY iii) ALIGN  
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Highlight the significance, features of Indirect addressing mode with suitable syntax or examples. (4)  
A) Course Outcome: "CO3" / Cognitive Level: "Application"  
B) Categorize the Load-Store instruction by suggesting few instructions used for that. (3)  
Course Outcome: "CO3" / Cognitive Level: "Analysis"  
C) List and illustrate different data processing instructions. (3)  
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) With a neat diagram, explain the working of a Barrel shifter by giving suitable examples. (4)  
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"  
B) Suggest the steps involved in case of fractional notation to evaluate any arithmetic function. (3)  
Course Outcome: "CO3" / Cognitive Level: "Application"  
C) With the syntax and examples, categorize the branching instructions used in ARM7. (3)  
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Discuss the significance of 'for' loop in ARM by illustrating the example of Summing integers. (4)  
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) Elaborate the concept of passing parameters by reference with example. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) Write the structure for ARM APCS specification for register usage. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) Highlight the significance of the following: i) ISR ii) Interrupt Vector and IVT. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) Categorize and explain the Quality attributes used in Embedded Systems. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Illustrate and explain an Application specific Embedded Systems. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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| Event Name | Subject |
|------------|---------|
|------------|---------|

058-MIT BLR-MAKEUP EXAM-2025-2026 ECM\_3122:Embedded Systems

| #            | Course Outcome | Description | Marks     |
|--------------|----------------|-------------|-----------|
| 1            | 1              | CO 1        | 4         |
| 2            | 2              | CO 2        | 6         |
| 3            | 3              | CO 3        | 20        |
| 4            | 4              | CO 4        | 10        |
| 5            | 5              | CO 5        | 10        |
| <b>Total</b> |                |             | <b>50</b> |

\* **NA** - CO Not Selected

| Event Name | Subject |
|------------|---------|
|------------|---------|

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| #            | Level       | No of Questions | Marks (%)        |
|--------------|-------------|-----------------|------------------|
| 1            | Remember    | 0               | 0(0%)            |
| 2            | Understand  | 3               | 10(20%)          |
| 3            | Application | 8               | 27(54%)          |
| 4            | Analysis    | 4               | 13(26%)          |
| 5            | Create      | 0               | 0(0%)            |
| 6            | Evaluate    | 0               | 0(0%)            |
| <b>Total</b> |             | <b>15</b>       | <b>50 (100%)</b> |